A canonical symmetry excludes the ramified quartic completion

Abstract

The fixed quartic selector is not universally adequate. A physical Alice–Bob transport of an already certified source lies outside its cone. Four explicit positive-definite integer block Hankel matrices separate this all-positive rational source. The transport also identifies the fractional Alice dependence missing from the present testing space. No larger family is asserted to cover the full canonical cone.

1 The counterexample and its validity

Retain the accepted source order $c = (d, \lambda, n, m, C, A, D, \alpha, \beta) \in \mathbb{R}^9_+$ and the complete canonical parent F_c. The cone $\mathcal{C}_{\mathrm{can}}$ consists of sources valid at every state and every pair of projective measurement angles. The accepted two-completion matrices are

$$G_J = \begin{pmatrix} P+K & J \\ J^T & N+K \end{pmatrix}, \qquad H_J = \begin{pmatrix} P+E & K-E-J \\ K-E-J^T & N+E \end{pmatrix}.$$

Here P, N, E, K are exactly the source-affine quadratic matrices in the canonical completion note [1]. The fixed cone $\mathcal{C}_{\mathrm{ram}}$ requires $v = 16s^2$, $s \geq 0$, and one matrix polynomial $J(s) = \sum_{k=0}^{4} J_k s^k$ making both matrices positive semidefinite. Its order-49 Gram pencil is retained unchanged.

The established input used below is the exact admission

$$\bar{c} = \left(1, 2^{-34}, 2^{-14}, 2^{-24}, \frac{1}{576}, 161, \frac{1}{2}, 76, \frac{1}{250}\right) \in \mathcal{C}_{\mathrm{ram}} \subseteq \mathcal{C}_{\mathrm{can}}. \tag{1}$$

Its existing proof and matrices are not recomputed.

Theorem 1. *The all-positive rational source*

$$\boxed{c^\dagger = \left(1, \frac{1}{576}, 161, 2^{-24}, 2^{-34}, 2^{-14}, \frac{1}{2}, \frac{1}{250}, 76\right)} \tag{2}$$

belongs to $\mathcal{C}_{\mathrm{can}} \setminus \mathcal{C}_{\mathrm{ram}}$. Consequently $\mathcal{C}_{\mathrm{ram}} \subsetneq \mathcal{C}_{\mathrm{can}}$: not every valid canonical source admits the fixed quartic selector. The accepted order-392 eight-copy outer lift is strictly larger than Q_{222}, by the established polar assembly theorem [2].

Define the involution

$$\pi c = (d, C, A, m, \lambda, n, D, \beta, \alpha), \qquad S(W, X, Y, Z) = (W, Y, X, Z).$$

The full parent satisfies the exact identity

$$\boxed{u^2 v^2 F_{\pi c}(-1/v, -1/u; S\psi) = F_c(u, v; \psi).} \tag{3}$$

For a direct verification, the eight event packets, in their accepted order $(\lambda, n, m, C, A, D, \alpha, \beta)$, are carried respectively to $(C, A, m, \lambda, n, D, \beta, \alpha)$. The m, D roots change sign. The d coefficient is unchanged: its three positive packets are permuted or fixed, and $D_u D_v Z^2$ is unchanged after clearing the angles. This proves (3) coefficient by coefficient, without a determinant argument.

There is no omitted endpoint. With homogeneous angle pairs $a = (a_0, a_1)$, $b = (b_0, b_1)$, identity (3) is the polynomial identity

$$\boxed{F^h_{\pi c}\big((b_1, -b_0), (a_1, -a_0); S\psi\big) = F^h_c(a, b; \psi).} \tag{4}$$

It holds on the complete product of angle circles, including literal zero coordinates. It permutes zero source weights exactly, keeps all state signs, and acts separately on source and testing coordinates. Real and imaginary state parts give the same conclusion for complex states. Thus $\pi \mathcal{C}_{\mathrm{can}} = \mathcal{C}_{\mathrm{can}}$. Since $c^\dagger = \pi \bar{c}$, (1) proves validity of (2). It remains to exclude its quartic completion.

2 An exact separating trace identity

For $\sigma = G, H$, let $N_{\sigma,0}, \dots, N_{\sigma,4}$ be the ten *integer* symmetric 4×4 matrices in Appendix A. Form the four integer block Hankel matrices

$$\mathsf{U}_\sigma = (N_{\sigma,i+j})_{i,j=0}^{2}, \qquad \mathsf{V}_\sigma = (N_{\sigma,i+j+1})_{i,j=0}^{1}. \tag{5}$$

They have orders $12, 8, 12, 8$ and are positive definite. Moreover,

$$(N_{G,k})_{\{1,2\},\{3,4\}} = (N_{H,k})_{\{1,2\},\{3,4\}} \qquad (0 \le k \le 4). \tag{6}$$

These properties are certified by the displayed integers and the exact elimination recurrence in the appendix.

Write $Q(s) = \sum_{k=0}^{4} Q_k s^k$ for either completed matrix. Its accepted half-line Gram representation is

$$Q(s) = Z_2(s)^T A_Q Z_2(s) + s Z_1(s)^T B_Q Z_1(s), \quad Z_2 = (I_4, sI_4, s^2 I_4)^T, \quad Z_1 = (I_4, sI_4)^T,$$

where $A_Q, B_Q \succeq 0$. Expanding gives

$$\sum_{k=0}^{4} \langle N_{\sigma,k}, Q_k \rangle = \langle \mathsf{U}_\sigma, A_Q \rangle + \langle \mathsf{V}_\sigma, B_Q \rangle. \tag{7}$$

Each J_k occurs in opposite off-diagonal blocks of G_k and H_k, so (6) cancels every selector coefficient. Consequently the sum of (7) is source-affine and nonnegative for every feasible certificate, including zero weights and singular Grams.

Here is the resulting integer functional in the accepted source order:

d	-97633735551856	λ	24154193783510329	n	85793705264
m	1787034292680750290	C	3303160925893294	A	16956553777510394
D	13682527570574	α	670221724556174	β	366431361010

(8)

Denote the functional with these coefficients by $\ell(c)$. Direct substitution of the accepted P, N, E, K, with $v = 16s^2$, gives the exact identity

$$\boxed{12\ell(c) = \langle \mathsf{U}_G, A_G \rangle + \langle \mathsf{V}_G, B_G \rangle + \langle \mathsf{U}_H, A_H \rangle + \langle \mathsf{V}_H, B_H \rangle.} \tag{9}$$

For checking its nine coefficients, only the moments of degrees $0, 2, 4$ are needed. If $Q_j^{(v)}(c, 0)$ is the coefficient of v^j with $J = 0$, the left side of (9) is

$$\sum_{j=0}^{2} 16^j \left[\langle N_{G,2j}, G_j^{(v)}(c,0) \rangle + \langle N_{H,2j}, H_j^{(v)}(c,0) \rangle \right]. \tag{10}$$

Thus $\ell \ge 0$ on $\mathcal{C}_{\mathrm{ram}}$. But direct rational evaluation yields

$$\boxed{\ell(c^\dagger) = -\frac{3260707096044735613817258 9}{9663676416000} < 0.} \tag{11}$$

This proves Theorem 1. No closure or dual-attainment argument is needed for this finite separating identity.

3 The contact mechanism and the remaining problem

The separation already holds at the exact contact source πc_* underlying (1). Its contact in positive angle gauges is

$$u = \frac{1}{16}, \qquad v = 13, \qquad \psi = \left(140, -\frac{1}{50}, \frac{1}{32}, 1\right),$$

and

$$\ell(\pi c_*) = -\frac{484433814365493288474548430934831 4410}{802811514300974151343419} < 0. \tag{12}$$

For clarity, both state-normal constraints are retained. Write

$$\eta = L(1/16)\psi = \left(\frac{71681}{8192}, \frac{17}{6400}, \frac{279}{32}, -\frac{801}{800}\right).$$

At $U = 1/256$ and $s = \sqrt{13}/4$, any putative completion must satisfy

$$G(s)(\eta_1, \eta_2, U\eta_3, U\eta_4)^T = 0, \qquad H(s)\eta^T = 0. \tag{13}$$

These follow from the accepted split identity and positivity of both summands at a zero. The separator excludes such a global quartic completion and remains negative after opening both zero event weights to the all-positive source (2).

The original proof's symmetry reduction preserves canonical validity, but it does not preserve the present fixed testing space. Its five compact Bob weights in affine coordinates are

$$\frac{(1, \sqrt{|v|}/4, |v|/16, |v|^{1/4}/2, |v|^{3/4}/8)}{\sqrt{1+v^2}}.$$

Under (3), they become the fixed Alice weights

$$\boxed{\frac{(|u|, \sqrt{|u|}/4, 1/16, |u|^{3/4}/2, |u|^{1/4}/8)}{\sqrt{1+u^2}}.} \tag{14}$$

Their branches and normalization depend only on the test. The homogeneous pullback of the accepted forty functions represents $b_0^2 F^h_{\pi c}$, where the original family represented $a_1^2 F^h_c$; both multipliers are nonnegative and positive on a dense testing chart.

The new dependence is genuinely outside the old span. Restrict to $u > 0$, $v = 1$, $\psi = (1, 0, 0, 0)$. Every old compact feature is a constant times a polynomial of degree at most three divided by $(1+u^2)^{3/2}$, and is analytic at $u = 0$. One transported feature is

$$-\frac{u^{1/4}}{16\sqrt{2}\sqrt{1+u^2}}.$$

It has nonintegral order of vanishing and cannot lie in that span. Over the common multiplier $a_1^2 b_0^2$, multiply old roots by b_0 and transported roots by a_1; the same feature becomes $-u^{5/4}/(16\sqrt{2}(1+u^2))$, so the distinction persists.

In particular $\mathcal{C}_{\mathrm{ram}}$ and $\pi\mathcal{C}_{\mathrm{ram}}$ are incomparable: (1), (2), and $\pi^2 = 1$ give a point in each set difference. Formula (14) is an exact transport of an entire certificate cone, but it does not prove that its addition covers $\mathcal{C}_{\mathrm{can}}$. Nor does separation prove that a particular quarter-power feature is the only possible repair.

Precise remaining implication. The implication $\mathcal{C}_{\mathrm{can}} \subseteq \mathcal{C}_{\mathrm{ram}}$ is false, rather than unresolved. The remaining task is to specify one finite, source-independent algebraic testing space $\mathcal{U}$ and prove that every $c \in \mathcal{C}_{\mathrm{can}}$ has its full unmultiplied slack, or one fixed testing-multiplied slack, in the Gram cone of $\mathcal{U}$. The R21 contact/conductor identities and qualitative tensor criterion [3] do not themselves identify that space. No such universal conversion is established here. The accepted companion, local certificates, and assembly results are unchanged.

Reproduction. Run `python3 verify.py`. The checker uses exact integer and rational arithmetic, verifies (4) for all nine coefficient packets, verifies (5)–(11) directly from the block definitions, and independently contracts the same certificate against all 148 auxiliary and nine source matrices of the unchanged public pencil. It also checks the value and all six first derivatives at the contact in (13). No numerical optimizer is required.

References

[1] *Canonical Bell validity through two matrix inequalities*; *An exact obstruction to the quadratic half-line completion*, and the accepted ramified order-49 data. Public repository, canonical-completion, commit 16b37f6.

[2] *Q222 Polar Assembly*, same commit, full-polar.

[3] Reconstructed R21 companion, same commit, latex, Chapters 3–6; in particular TF12, SG2, SG7, and CC1–CC6.

A The complete integer moment certificate

All matrices below are used as integers in (5)–(10). The machine-readable file also records the common normalization $N/(15 \cdot 2^{60})$; multiplying a dual certificate by a positive constant does not change its sign or its cancellations.

Positive definiteness can be checked without numerical eigenvalues. For any matrix R in (5), use the rational recurrence

$$p_j = R_{jj} - \sum_{k<j} L_{jk}^2 p_k, \qquad L_{ij} = \frac{R_{ij} - \sum_{k<j} L_{ik} p_k L_{jk}}{p_j}, \quad L_{jj} = 1 \quad (i > j).$$

It gives $R = L\,\mathrm{diag}(p_j)L^T$. The exact floors of the smallest pivots are

R	U_G	V_G	U_H	V_H
$\lfloor \min_j p_j \rfloor$	794	12397	67408	1527738.

Thus all four matrices are positive definite. The independent checker instead uses fraction-free leading principal determinants and obtains the same pivot bounds. The shared upper-right blocks in each pair $N_{G,k}, N_{H,k}$ make (6) directly visible.

$$N_{G,0} = \begin{pmatrix} 72283584904157745 & 98314305759870 & 214652490042795 & -146888491666410 \\ 98314305759870 & 135542238435 & 268778302935 & -201769656060 \\ 214652490042795 & 268778302935 & 1297549019460 & -401316089475 \\ -146888491666410 & -201769656060 & -401316089475 & 301979389605 \end{pmatrix}.$$

$$N_{G,1} = \begin{pmatrix} 29553765112790565 & 39441163448385 & 105424012102020 & -58918574211480 \\ 39441163448385 & 54072935055 & 110357620980 & -80768779620 \\ 105424012102020 & 110357620980 & 1079281903875 & -164509196205 \\ -58918574211480 & -80768779620 & -164509196205 & 120695415960 \end{pmatrix}.$$

$$N_{G,2} = \begin{pmatrix} 12789595044635020 & 15916490027610 & 66271058811705 & -23763057906660 \\ 15916490027610 & 21653142535 & 47712869220 & -32349910290 \\ 66271058811705 & 47712869220 & 1287764254705 & -70816288035 \\ -23763057906660 & -32349910290 & -70816288035 & 48333328645 \end{pmatrix}.$$

$$N_{G,3} = \begin{pmatrix} 6842765115107160 & 6582732635010 & 46453400242755 & -9802159230735 \\ 6582732635010 & 8693049165 & 22782075435 & -12984692925 \\ 46453400242755 & 22782075435 & 1981266595425 & -33639345030 \\ -9802159230735 & -12984692925 & -33639345030 & 19395668925 \end{pmatrix}.$$

$$N_{G,4} = \begin{pmatrix} 20910075570761184 & 4127281354275 & 33265473661770 & -6066895933245 \\ 4127281354275 & 3608024196 & 12294756105 & -5380775115 \\ 33265473661770 & 12294756105 & 5215970784456 & -19334942190 \\ -6066895933245 & -5380775115 & -19334942190 & 8026177596 \end{pmatrix}.$$

The five H moments

$$N_{H,0} = \begin{pmatrix} 51123677524583550 & -5656910689200 & 214652490042795 & -146888491666410 \\ -5656910689200 & 10186908915 & 268778302935 & -201769656060 \\ 214652490042795 & 268778302935 & 146089272033990 & -170629846948365 \\ -146888491666410 & -201769656060 & -170629846948365 & 2302846743948255 \end{pmatrix}.$$

$$N_{H,1} = \begin{pmatrix} 765887260600020 & 39305335635 & 105424012102020 & -58918574211480 \\ 39305335635 & 2703893715 & 110357620980 & -80768779620 \\ 105424012102020 & 110357620980 & 92030582539560 & -33880347103335 \\ -58918574211480 & -80768779620 & -33880347103335 & 160298762935095 \end{pmatrix}.$$

$$N_{H,2} = \begin{pmatrix} 89446398381610 & 50022371745 & 66271058811705 & -23763057906660 \\ 50022371745 & 931545700 & 47712869220 & -32349910290 \\ 66271058811705 & 47712869220 & 64202652830605 & -13709279518560 \\ -23763057906660 & -32349910290 & -13709279518560 & 18313581996550 \end{pmatrix}.$$

$$N_{H,3} = \begin{pmatrix} 48316110408750 & 24625867005 & 46453400242755 & -9802159230735 \\ 24625867005 & 328684920 & 22782075435 & -12984692925 \\ 46453400242755 & 22782075435 & 45819719138655 & -8466528163695 \\ -9802159230735 & -12984692925 & -8466528163695 & 3366601507110 \end{pmatrix}.$$

$$N_{H,4} = \begin{pmatrix} 33727791386796 & 13046747130 & 33265473661770 & -6066895933245 \\ 13046747130 & 116838834 & 12294756105 & -5380775115 \\ 33265473661770 & 12294756105 & 32917216546116 & -5858691614355 \\ -6066895933245 & -5380775115 & -5858691614355 & 1299098148456 \end{pmatrix}.$$